Post-impact fatigue performance of carbon fibre laminates with non-woven and mixed-woven layers

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Woven and non-woven carbon fibre laminates were subjected to drop-weight impact at energies of up to seven Joules. The damaged laminates were tested to differing periods of O-tension and O-compression fatigue, and then static residual strength tests were performed. The non-woven composite showed greater losses in static strength after impact than the mixed-woven composite, particularly when tested in compression. After tensile fatigue, significant improvements in residual strength were recorded, which coincided with the observation of micro-delaminations throughout the laminate. After compressive fatigue there was little change in residual strength.

Key words: *composite materials; drop tests; fatigue testing; residual strength; woven fabrics; carbon fibres; epoxy resins; laminates.*

With the recent growth in the application of carbon fibre composites in aircraft structures, considerable attention is being focussed at understanding the effect of impact damage on such materials.

Low velocity impact tests on CFRP have shown[1] that considerable delamination and cracking is incurred under subsequent dynamic loading. Work by Dorey *et al*[2] considered the importance of the specimen geometry in determining its impact resistance. It was concluded that the span to depth ratio (s/d) of a beam determines the nature of damage incurred. Low values of s/d tend to promote longitudinal splitting whereas larger values encourage transverse cracking. It was concluded that when designing for higher threshold energies for the onset of damage, a span to depth ratio that favours simultaneous splitting and delamination should be chosen.

Low velocity impact tests[3] have indicated that the post-impact flexural properties are very susceptible to the presence of transverse cracking on the lower face of the specimen. The compressive properties of CFRP have been found to be sensitive to delamination-type damage.[4]

The post-impact fatigue properties of carbon fibre composites have also received considerable attention.[3,5] Fatigue tests on a 16 ply multidirectional composite[5] led to improvements of up to 40% in the static tensile strength and 25% in the static compression strength. It was suggested that this effect was due to the extension of microcracks within the laminate which produced a larger damage zone and consequently a greater failure stress.

The composite stacking sequence has been shown to strongly influence both the impact resistance and the subsequent fatigue resistance of the laminate.[5] Further tests[3] indicated that the extent of the impact damage can be reduced by the inclusion of 45° plies on the outermost surfaces of a composite, which protect the main load carrying 0° fibres.

The role of the resin in the impact behaviour of composites is complex. There are a number of energy absorbing mechanisms such as fibre pull-out, fibre fracture, delamination *etc* which are affected by the resin properties.

Rogers[6] found that increasing the strain to failure of the resin improved the impact resistance of composites.

This report considers the possible advantages in impact performances of employing a woven fabric in place of some of the individual laminate plies. The effect of O-tension and O-compression fatigue on the post-impact performance of these laminates is considered.

EXPERIMENTAL PROCEDURE

The laminates studied were manufactured from 0.125 mm thick pre-impregnated sheets of high strength, surface-treated carbon fibres in an epoxy resin matrix layed-up in the following manner:

$$[+45, -45, 0, 0, -45, +45, 0, 0]_s$$

This gave panels with a nominal thickness of 2 mm and a volume fraction of 60%. Similar laminates were manufactured using a ±45° woven fabric in place of the individual 45° plies, resulting in slightly thicker panels.

After manufacture, the laminates were scanned using an ultrasonic probe immersed in water to assess their quality and to determine areas of potential weakness.

Impact testing was undertaken using a standard drop-weight test rig with a 12.7 mm spherically nosed weight falling through a height of one metre to strike the panels centrally. The panels were clamped betwéen two rings of 100 mm internal diameter; this method of support was thought to simulate practical impact conditions. By changing the mass of the falling carriage, the incident impacting energy was varied between 1 and 7 J. After impact, the carriage was caught to avoid secondary impact.

The impact damaged panels were scanned ultrasonically and examined visually to assess the type and extent of damage sustained and then cut into individual coupons of dimensions 250 x 50 x 2 mm for further testing.

Aluminium tags 50 x 50 x 2 mm were glued on to the ends of all specimens designated for static and fatigue tests in tension. The static tensile tests were performed on an Instron Universal testing machine with a crosshead speed of 2 mm/min.

Quasi-static tensile and compressive tests were conducted on specimens corresponding to each damage level to determine the limits required for fatigue. In compression testing, an anti-buckling guide was used to support the specimen edges in a manner which allowed free longitudinal movement. Fatigue cycling was conducted at 80% of the average residual static strengths. The fatigue testing was carried out on a Lowsenhausen testing machine in a load control mode with a cycling frequency of 20 Hz.

Cycling was interrupted after 10^4, 10^5 or 10^6 reversals and coupons scanned to assess the growth of damage. The residual static strengths were then measured.

RESULTS

Drop-weight impact was found to initiate considerable visible cracking and delamination, particularly at impact energies of 6 J and 7 J. The damage after an impact of 7 J was so extensive that it propagated into areas of the panel from which neighbouring specimens were to be taken. To eliminate this effect the maximum impact energy was set at 6 J.

Fig. 1 shows the C-scans of both woven and non-woven specimens subjected to varying impact energies. It should be noted that the C-scan technique primarily shows areas of delamination. No damage was apparent in the woven and non-woven specimens, either visually or ultrasonically, after being subjected to an impact energy of 1 J. The delamination in the non-woven specimens grew in the direction of the outer 45° ply and considerable peeling of this outer layer was observed at an impact energy of 6 J. Delamination in the woven coupons was less extensive than in the non-woven specimens, in this case the damage propagated mainly in the 0° direction.

Fig. 2 shows the C-scans of those laminates subjected to varying degrees of O-tension and O-compression fatigue. Included for comparison are the corresponding scans of the coupons prior to cycling.

The non-woven laminate did not exhibit an increase in delaminated area before 10^5 cycles O-tension fatigue, but a 100% increase was observed after 10^6 reversals, the damage having propagated in the direction of the applied loading. Rather surprisingly, no increase in the delamination zone size was noted after 10^6 cycles O-compression fatigue. Work by Card and Rhodes[7] suggests that the compressive fatigue properties of a beam are very susceptible to delamination-type damage. Micrographs of a similar lay-up before and after O-compression fatigue are shown in Fig. 3. In both of these cases no increase in the ultrasonic damage was noted, but Fig. 3b does indicate an increased level of damage.

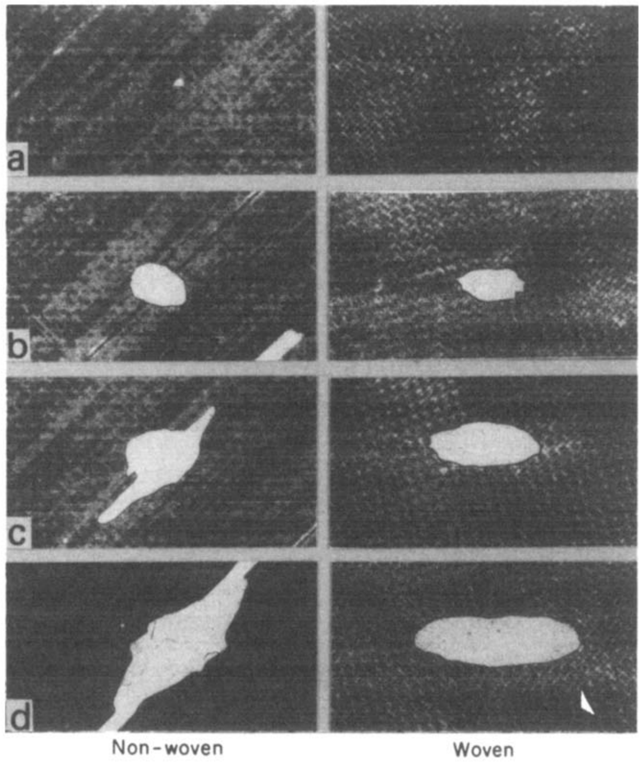

Fig. 1 Ultrasonic C-scans of non-woven and mixed-woven laminates after low velocity impact at various incident energies: (a) 1 J; (b) 2 J; (c) 4 J; and (d) 6 J

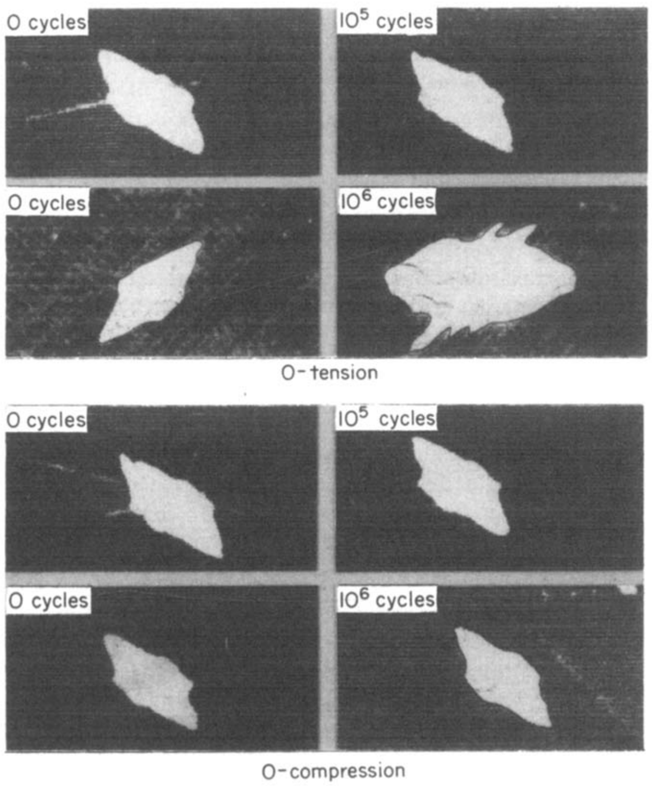

Fig. 2 Ultrasonic C-scans of non-woven laminates subjected to an impact energy of 4 J and subsequently fatigued in O-tension or O-compression

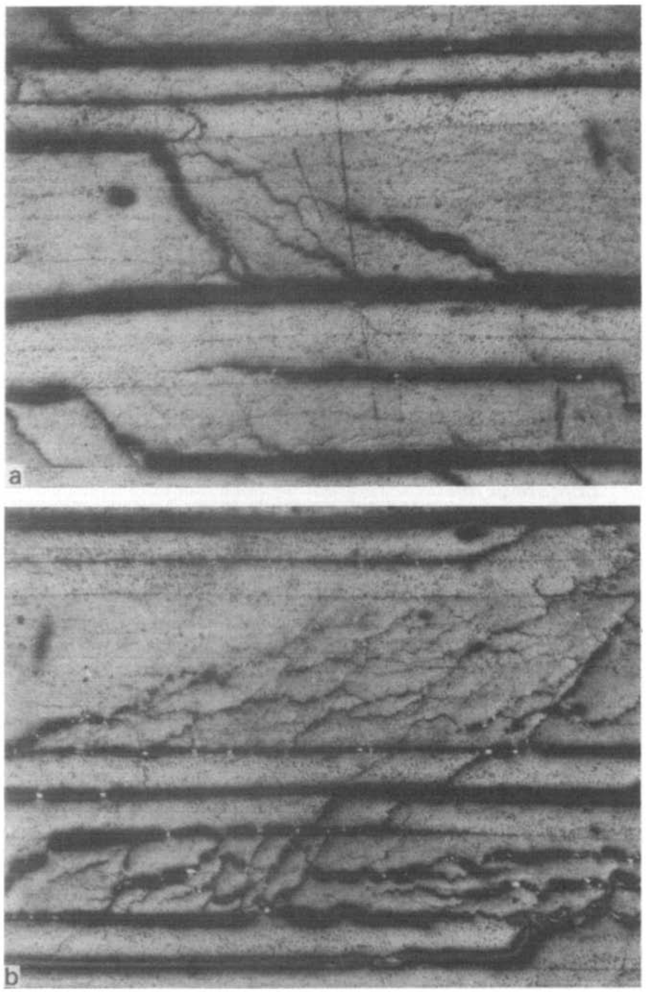

Fig. 3 Micrographs of non-woven impact damaged laminates before and after 10^5 cycles of O-compression fatigue. The impact energy was 4J

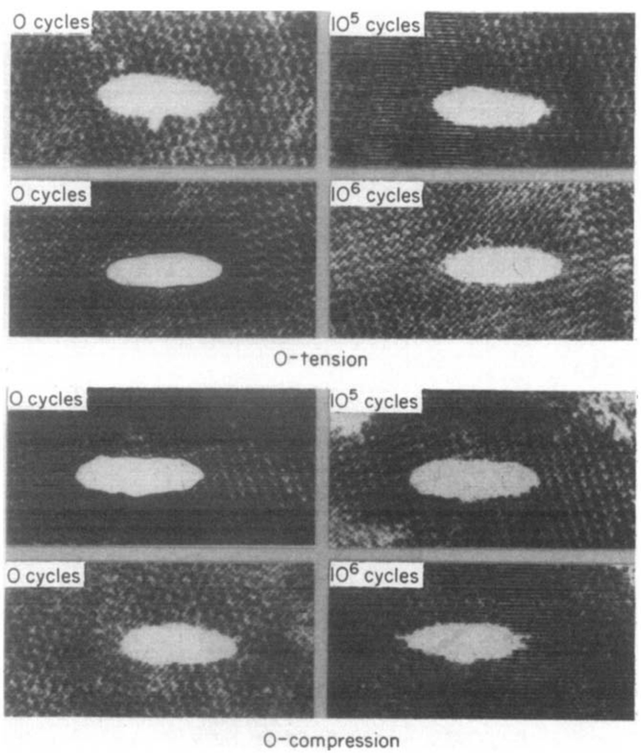

Fig. 4 Ultrasonic C-scans of mixed-woven laminates subjected to an impact energy of 4 J and subsequently fatigued in O-tension or O-compression

The scans of the woven specimens subjected to an impact energy of 4 J are shown in Fig. 4. No increase in the damage zone size was apparent in either O-tension or O-compression fatigue. However, the woven coupon subjected to an impact energy of 2 J and 10^6 cycles O-tension fatigue did exhibit signs of damage initiation and growth. The C-scan of this specimen showed an overall lightening, which was later attributed to the presence of micro-delaminations within the ±45° woven fabric.

The cycled and uncycled residual strength curves are shown in Figs 5–8.

DISCUSSION

Extensive damage was incurred during drop-weight impact at energies up to 7 J. Micrographs of impact damaged specimens revealed considerable through-the-thickness cracking and delamination.

Fig. 3 verifies the presence of shear cracking near the point of impact; these cracks tend to radiate from the point of impact in a fashion similar to that noted by Card and Rhodes.[7] Considerable delamination was also apparent, particularly between the ±45° plies, *ie* between plies at which the greatest change of fibre direction occurred. This tendency was also noted by Card and Rhodes.[7] The type and consequence of the damage discussed previously had differing effects on the post-impact static performance of the laminates considered and is discussed below.

Non-woven laminate

1. Tension

The non-woven test specimens clearly exhibited a sensitivity to impact damage. Fig. 5 shows a 50% reduction in static tensile strength occurs over the given energy range.

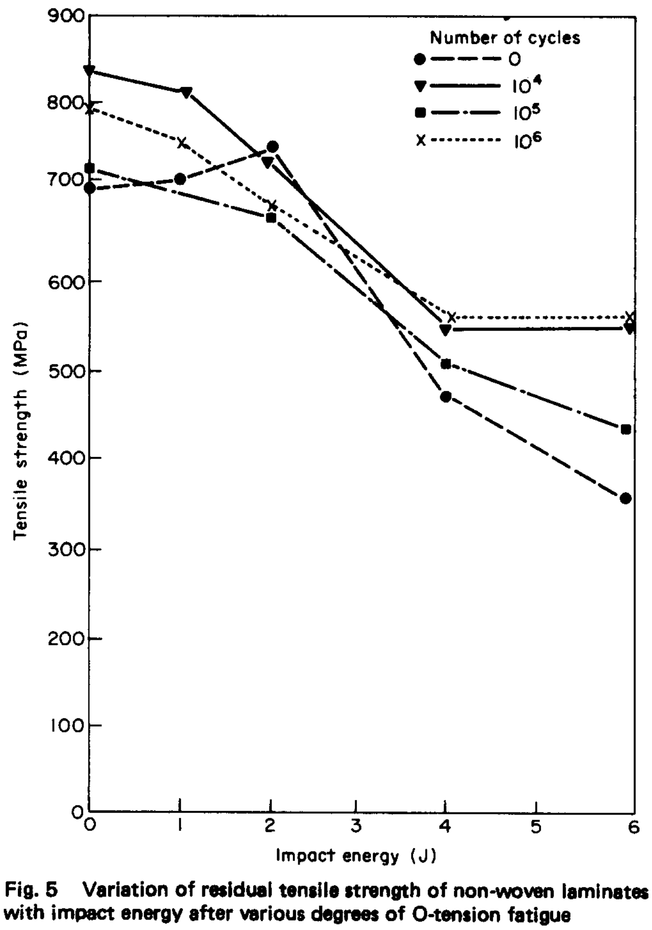

Fig. 5 Variation of residual tensile strength of non-woven laminates with impact energy after various degrees of O-tension fatigue

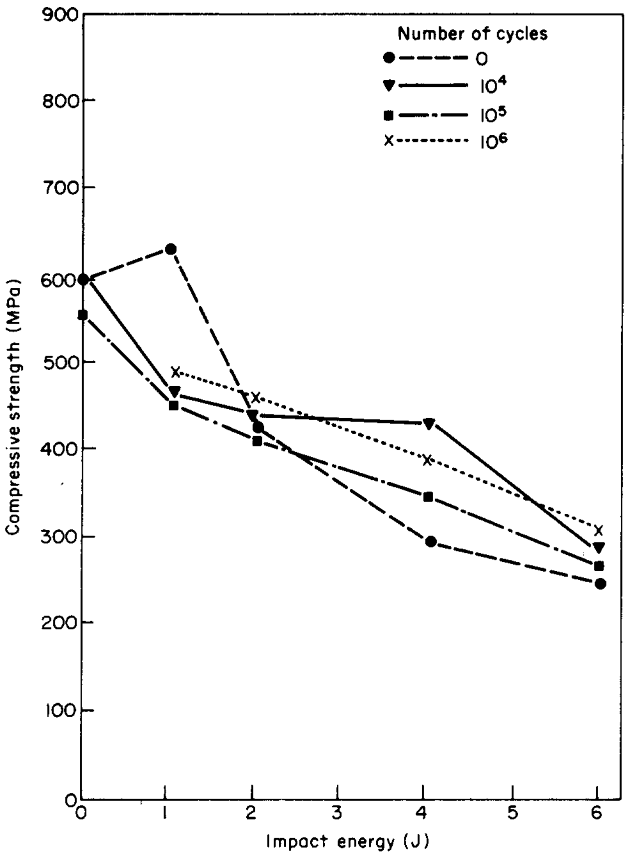

Fig. 6 Variation of residual compressive strength of non-woven laminates with impact energy after various degrees of O-compression fatigue

Fig. 5 also indicates that cycling this laminate yields improved static strengths, most particularly at high impact energies. It is interesting to note that the specimen subjected to an impact energy of 4 J and 10^6 cycles O-tension fatigue exhibited a 20% increase in static tensile strength even though the C-scans indicated a 100% increase in delaminated area (Fig. 2).

The reason for this improved static performance following cycling is not clear, although it could be associated with a notch blunting mechanism similar to that noted by Bishop and Morton[8] during tests on similar notched laminates.

2. Compression

Fig. 6 shows that the compressive properties of this non-woven laminate were clearly sensitive to impact damage. A marked loss in static strength was evident at high impact energies, which was due to the ease of fibre microbuckling within the delaminated region.

Cycling the laminate under O-compression fatigue resulted in increased residual strengths and a considerably more linear curve.

Woven laminates

1. Tension

The most striking feature of the residual strength curves shown in Fig. 7 is the extent of the scatter in the results. The static curve displayed a sharp increase in strength between 4 J and 6 J. This was due to an unusually large delaminated region in the 4 J specimen, which was probably due to an initial flaw within the specimen. The loss in static strength (approximately 20%) over the energy range considered was much less than in the non-woven laminate, as has been observed elsewhere.[9] This was further reduced with continued cycling; indeed those specimens subjected to 10^6 cycles exhibited similar static strengths regardless of the initial impact energy.

The coupon subjected to an impact energy of 2 J and then fatigued for 10^6 cycles showed extensive micro-delaminations throughout the ±45° woven fabric. This delamination initiated as a result of the nature of the weave, a five shaft satin weave (four over, one under fabric).

This extensive damage did not, however, affect the static strength of the specimen; indeed this coupon was stronger than its uncycled counterpart. Since the effect was not apparent at greater energy levels, a critical cycling stress level for the formation of this damage must exist.

2. Compression

This laminate exhibited reduced residual strengths after sustaining O-compression fatigue. Although no increase in delaminated area was noted, Fig. 8 shows that the static strength was reduced by up to 20% after cycling. There was no indication of the initiation and propagation of the micro-delaminations observed in the tensile testing. It must therefore be concluded that any damage initiated during compressive cycling on this lay-up must be too small to be detected ultrasonically.

The tests indicate that this laminate may be sensitive to O-compression fatigue and might possibly fail with continued cycling.

CONCLUSIONS

Both the non-woven and the mixed-woven laminates exhibited a reduction in strength as a result of low velocity

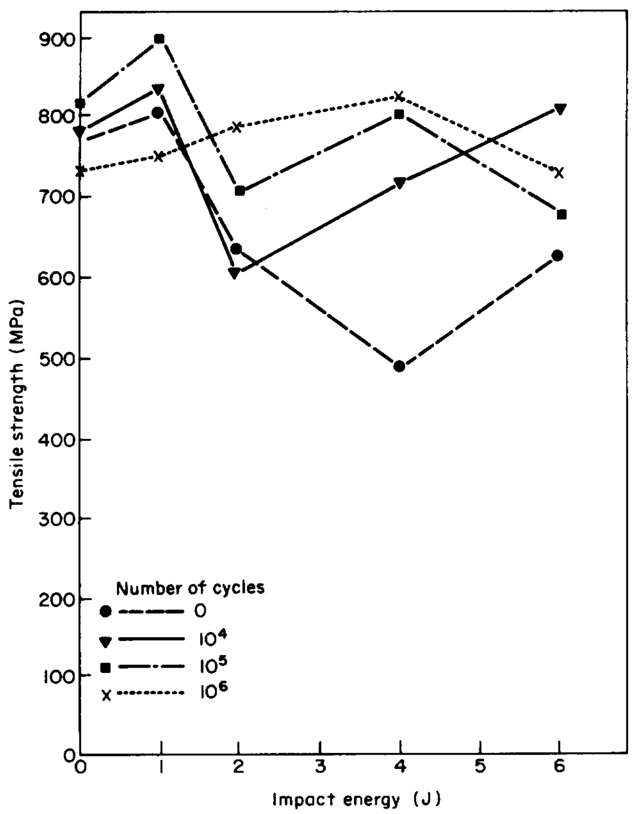

Fig. 7 Variation of residual tensile strength of mixed-woven laminates with impact energy after various degrees of O-tension fatigue

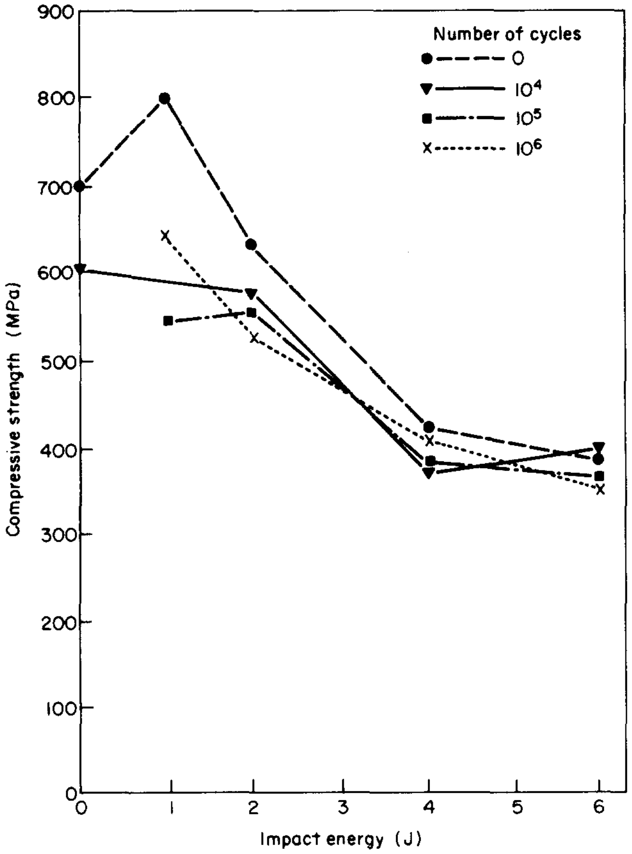

Fig. 8 Variation of residual compressive strength of mixed-woven laminates with impact energy after various degrees of O-compression fatigue

impact loading. The residual tensile strengths of the mixed-woven laminates were significantly greater than those of the non-woven laminates. In both laminates, O-tension fatigue cycles improved the residual strengths for a given incident impact energy. In the case of the mixed-woven composites, the post-fatigue residual tensile strengths showed little dependence upon the incident impact energy.

The residual compressive strengths showed a marked reduction with increasing impact energy, the effect being greater in the case of the non-woven laminates. After impact with an incident energy of 6 J, the residual compressive strength of the non-woven laminate was less than 50% of the undamaged compressive strength. The effect of post-impact fatigue in the non-woven composite tended to increase the residual strength. However, this effect was less apparent in the case of mixed-woven laminates in which, for lower impact energies at least, fatigue cycles resulted in further reductions in the residual compressive strengths.

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